Experiment 29
Fraunhofer diffraction

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Abstract

This experiment explores the principles of Fraunhofer diffraction by analyzing the diffraction patterns produced when monochromatic laser beams pass through single slits and diffraction gratings. Using red ($\lambda \approx 730nm$) and green ($\lambda \approx 532nm$) lasers, we first determined the width of various slits by measuring the angular spread of the central diffraction maximum. In the second part, we reversed the method to calculate the wavelengths of the lasers by using known diffraction grating . The results revealed a discrepancy between the measured and expected wavelengths, suggesting instrumental errors. However, the experiment successfully demonstrated the application of Fraunhofer diffraction.

Contents

1 Introduction **1**
1.1 Contextualization . . . 1
1.2 Theorical principles . . . 2
1.2.1 Diffraction by a Slit . . . 2
1.2.2 Diffraction by a Diffraction Grating . . . 2

2 Materials and Methods **3**
2.1 Materials . . . 3
2.2 Methods . . . 4
2.2.1 Variable-width slit . . . 4
2.2.2 Determination of the wavelength of a laser beam . . . 5

3 Results and discussion **6**
3.1 Slit width measurement . . . 6
3.2 Determination of the wavelength of a laser beam . . . 6

4 Conclusions **7**

5 Appendixes **7**
5.1 Calculation of Uncertainties . . . 7

1 Introduction

1.1 Contextualization

Diffraction is a cornerstone of wave optics, demonstrating light's deviation from rectilinear propagation when interacting with obstacles or apertures. First quantitatively described by

Francesco Grimaldi in the 17th century, diffraction underpins modern technologies such as optical spectroscopy, X-ray crystallography, and laser applications.

Fraunhofer diffraction, a specific regime where the observation screen is placed far from the diffracting object, simplifies analysis by eliminating near-field complexities [1].

This experiment examines Fraunhofer patterns produced by laser light passing through single slits and diffraction gratings, enabling precise determination of light wavelengths and slit dimensions—a technique widely used in material science and telecommunications [2].

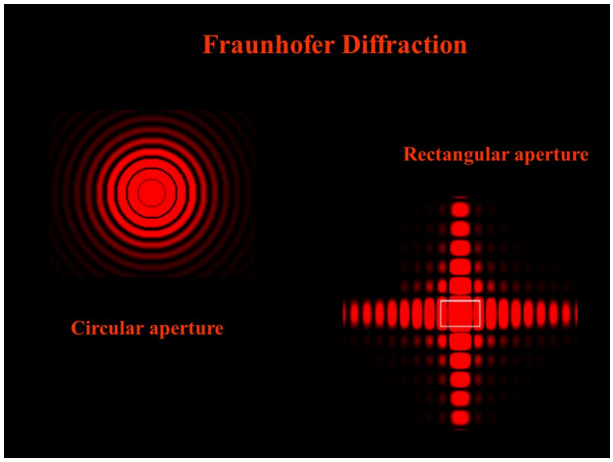


Figure 1: Circular and rectangular figures we can see analyzing Fraunhofer patterns.

1.2 Theorical principles

1.2.1 Diffraction by a Slit

When monochromatic light passes through a single slit of width a, it produces a diffraction pattern with a central bright fringe surrounded by alternating dark and bright fringes. The angular width θ of the central fringe is given by

$$\theta = \frac{2\lambda}{a}, \tag{1}$$

where λ is the wavelength of light. This relationship allows for experimental determination of either the slit width or the light wavelength when the other is known.

1.2.2 Diffraction by a Diffraction Grating

A diffraction grating consisting of multiple equally spaced slits (spacing d) is produced by the interference of the interference between the diffracted waves emerging from each slit.

The angle of diffraction θ determines the interference pattern, with intensity peaks appearing when the phase difference between waves corresponds to integer wavelength multiples:

$$d \sin \theta_m = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots). \tag{2}$$

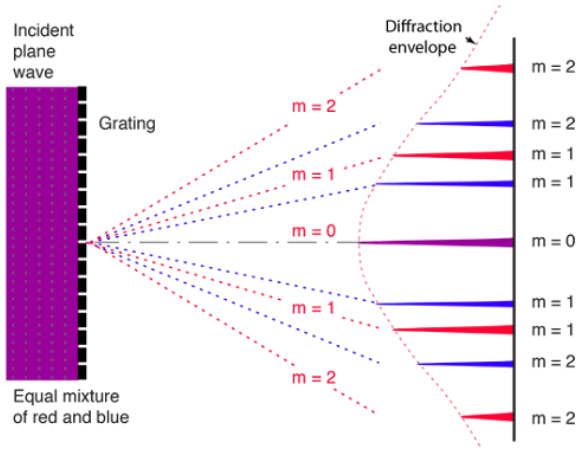


Figure 2: Diffraction by a diffraction grating with m values represented.

It is important to noctice thart the angular separation between maxima increases with:

1. Longer wavelengths (λ)
2. Smaller grating spacing (d)
3. Higher diffraction order (m)

Experimentally, we can determinate θ for a given maximum intensity of order m and calculate its sine as

$$sin(\theta) = \frac{D}{\sqrt{D^2 + L^2}}. \tag{3}$$

The position of these maxima on a screen at distance L can be used to calculate the wavelength of light clearing λ in equation 2 and substituting with equation 3:

$$\lambda = \frac{d \cdot D}{\sqrt{D^2 + L^2}}, \tag{4}$$

where D is the distance from the central maximum to the m-order maximum.

2 Materials and Methods

2.1 Materials

The materials used in this practice are the following[3].

- **Laser device**. Beam of two different wavelengths, red ($\lambda_r = 635nm$) and green ($\lambda_g = 532nm$).
- **Diffraction sheets**. Two distinct plates, one with three slits of different widths (A, B and C); and the other one with three gratings of 944.88, 2952.8 and 5905.5 lines per centimeter.
- **Screen**, in which we will be able to see the different diffraction patterns.
- **Optical bench**. Instrument where we will position the laser, diffraction sheets and screen.
- **Graph paper**, employed to size the gap between two consecutive maximums of the diffraction pattern. This utensil has a resolution of 0,001 m.
- **Measuring tape**. Used to measure the distance between the screen and the sheet with a precision of 0,01 m.



Figure 3: Red laser crossing the thinner slit, i.e., the slit A.

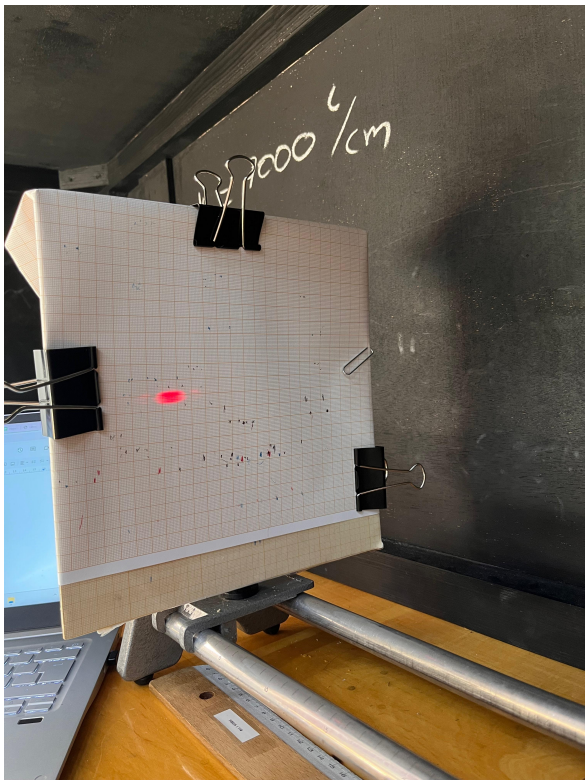


Figure 4: Diffraction pattern of the situation described in 3.

2.2 Methods

2.2.1 Variable-width slit

The main objective of this part of the experiment is to determine the width of each slit. To do so, we will follow the steps explained in [3].

1. Firstly, we will shine the laser beam on each of the three slits with with the sheet in a fixed position and measure the distance x between this point and the screen, as well as the width h of the central strip formed after the ray goes through the slit.

2. We will repeat the first step changing the position of the screen thrice (adding 10 cm each time), and with both colours, obtaining eighteen measurements. The data will be organized in a table for future analysis.

3. To determine the angle θ, necessary to obtain the width a of the slits, we will use equation 5.

4. Lastly, we will calculate a applying the formula 1

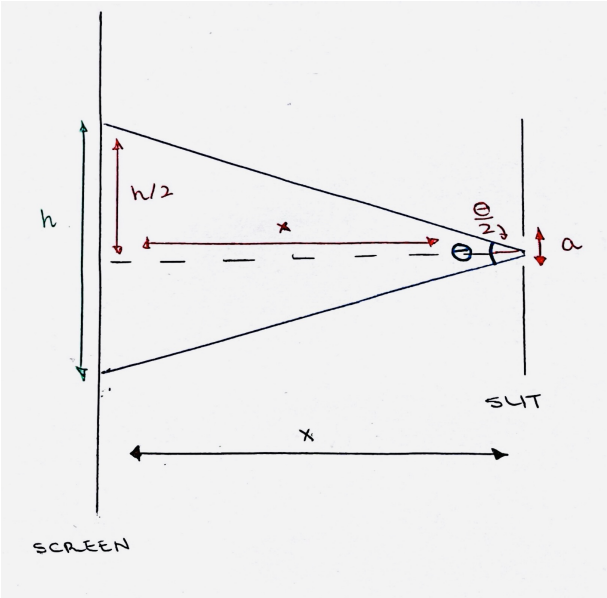


Figure 5: Diffraction by a slit

From this sketch it is clear that the angle θ is defined as:

$$\theta = 2arctan\left(\frac{h}{2x}\right) \tag{5}$$

2.2.2 Determination of the wavelength of a laser beam

In this section of the experience we will determine the wavelength of each colour. In order to achieve this, we will follow the next procedure, as stated in [3].

1. In the first place, we need to calculate the constant k in lines/m, and, with this value, obtain d, defined as follows.
$$d = \frac{1}{k}(m^{-1}) \tag{6}$$

2. Then, we will direct the laser beam on each of the three gratings with the sheet in a fixed position and measure the distance L between this point and the screen, as well as the gap D between the central maximum -the one that shines the most- and the next one, which emerge on the screen after the ray crosses through the diffraction grating.

3. We will repeat the second step changing the position of the screen thrice (adding 10 cm each time), and with both colours, obtaining eighteen measurements. The collected data will be arranged in a table for later examination

4. Lastly, λ will be determined for each colour using equation 4 The average value $\overline{\lambda}$ will be compared with the one given by the manufacturer.

3 Results and discussion

3.1 Slit width measurement

In the first part of the experiment, we calculated the width a for three different slits (A, B, and C) by analyzing the diffraction pattern produced by both red and green laser beams. The wavelengths of the lasers used were $\lambda = 532$ nm (green) and $\lambda = 650$ nm (red). The measured values are summarized in Table 1.

Slit	Red laser's width a (μm)	Green laser's width a (μm)
A	$140,44 \pm 1,41$	$115,1 \pm 1,2$
B	$306,04 \pm 6,74$	$261,884 \pm 6,023$
C	$544,91 \pm 21,71$	$414,96 \pm 15,54$

Table 1: Width a of A, B and C, calculated for red and green laser with corresponding uncertainties

The results show that the measured slit widths using the green laser are consistently smaller than those obtained using the red laser. This difference is likely due to the shorter wavelength of the green laser, which leads to a broader diffraction pattern for the same slit width, thus increasing the sensitivity of the measurement. As such, small errors in fringe position can cause more pronounced deviations in the calculated width for longer wavelengths.

As can be seen, the uncertainties obtained for the width of each slit are quite small, which indicates that we can be certain of the precision of our results.

3.2 Determination of the wavelength of a laser beam

In the second part of the experiment, we reversed the approach and used known gradings to determine the wavelengths of the red and green lasers. The results obtained are shown in the following table.

Laser	Wavelength (λ) (nm)
Green laser	$586,4 \pm 2,1$
Red laser	$678,62 \pm 0,28$

Table 2: Red and green laser's wavelength expressed in nanometers and their corresponding uncertainties.

The wavelength obtained for the green laser should ideally fall within the range of 491 nm to 568 nm, according to the reference values provided in [4], which correspond to the visible spectrum of green light typically used in commercial laser pointers and laboratory equipment. Similarly, the red laser is expected to have a wavelength around 730 nm, which places it in the deep red region of the visible spectrum.

However, as shown in Table 2, our experimentally determined wavelengths deviate from these expected values. We measured the green laser to have a wavelength of 586.4 ± 2.1 nm, which places it closer to the yellow-orange region, and the red laser at 678.62 ± 0.28 nm.

These discrepancies suggest the presence of systematic errors in the experimental procedure or limitations in the accuracy of the measurement setup. One possible cause is the low resolution of the instruments used compared to the distances we needed to measure.

4 Conclusions

The Fraunhofer diffraction experiment successfully demonstrated the principles of diffraction through single slits and diffraction gratings. In the first part of the experiment, we measured the slit widths for both red (650 nm) and green (532 nm) lasers by analyzing the diffraction patterns. The results showed that smaller slit widths produced broader diffraction patterns, and the green laser, with its shorter wavelength, exhibited a more sensitive response, which resulted in smaller slit width measurements compared to the red laser.

In the second part of the experiment, we used known diffraction gratings to calculate the wavelengths of the lasers. However, the experimentally determined wavelengths showed a significant discrepancy from the expected values, with the green laser measured at 586.4 ± 2.1 nm (instead of approximately 532 nm) and the red laser at 678.62 ± 0.28 nm (rather than 730 nm). This discrepancy suggests potential systematic errors, possibly due to instrumental limitations.

Despite these errors, the experiment successfully demonstrated the application of Fraunhofer diffraction as a method for analyzing light properties. The observed diffraction patterns, in line with theoretical predictions, reinforced the wave nature of light and its interaction with apertures.

5 Appendixes

5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{7}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations.

On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{8}$$

where δ_x is the resolution of the measuring instrument (δ_h=0,001 m)

From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{9}$$

Additionally, we can calculate the expanded uncertainty is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor kp that is selected for convenience:

$$U_C(x) = u_C(x)k_p. \tag{10}$$

This factor is derived from the distribution of the t-student, using $n-1$ degrees of freedom and a fixed confidence level. In this case, consistently 95%:

$$kp = 2.178813.$$

With these equations we are able to characterize the uncertainties for every variable used. For the distances x and L between the screen and the sheet, we get:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)} = u_B(x) = \frac{0,01}{\sqrt{12}} = 0,0029(m) = u_C(L),$$

and for the calculus of h and D, which we defined as the width of the central strip and the distance between two consecutive maximums respectively,

$$u_C(h) = \sqrt{u_A^2(h) + u_B^2(h)} = u_B(h) = \frac{0,001}{\sqrt{12}} = 0,00029(m) = u_C(D).$$

Regarding the error of the calculated θ, since it is an indirect measurement obtained from equation 5, we express:

$$u_C(\theta) = \sqrt{\left(\frac{\partial\theta}{\partial x}\right)^2 (u_C(x))^2 + \left(\frac{\partial\theta}{\partial h}\right)^2 (u_C(h))^2} \rightarrow$$

$$\rightarrow u_C(\theta) = \sqrt{\left(\frac{-4h}{4x^2+h^2}\right)^2 (u_C(x)^2) + \left(\frac{4x}{4x^2+h^2}\right)^2 (u_C(h)^2)(^\circ)}. \tag{11}$$

To calculate the error associated to the width of each slit, a, we use the following equation:

$$u_C(a) = \sqrt{\left(\frac{\partial a}{\partial\theta}\right)^2 (u_C(\theta))^2 + \left(\frac{\partial a}{\partial\lambda}\right)^2 (u_C(\lambda))^2} = \sqrt{\left(\frac{-2\lambda}{\theta^2}\right)^2 (u_C(\theta)^2)}(m), \tag{12}$$

and,to obtain the error of the wavelength λ we use:

$$u_C(\lambda) = \sqrt{\left(\frac{\partial\lambda}{\partial L}\right)^2 (u_C(L))^2 + \left(\frac{\partial\lambda}{\partial D}\right)^2 (u_C(D))^2 + \left(\frac{\partial\lambda}{\partial d}\right)^2 (u_C(d))^2} \rightarrow$$

$$\rightarrow u_C(\lambda) = \sqrt{\left(\frac{-dDL}{(D^2+L^2)^{\frac{3}{2}}}\right)^2 (u_C L))^2 + \left(\frac{dL^2}{(D^2+L^2)^{\frac{3}{2}}}\right)^2 (u_C(D))^2}(m). \tag{13}$$

Finally, for every average value calculated we use the following fromula.

$$u_C(\overline{x}) = \sqrt{\left(\frac{\partial\overline{x}}{\partial x_1}\right)^2 (u_C(x_1))^2 + ... + \left(\frac{\partial\overline{x}}{\partial x_n}\right)^2 (u_C(x_n))^2} = \frac{1}{n}\sqrt{(u_C(x_1))^2 + ... + (u_C(x_n))^2}.$$

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[4] HedUE. "Colores láser". In: *Hedue* (2025).